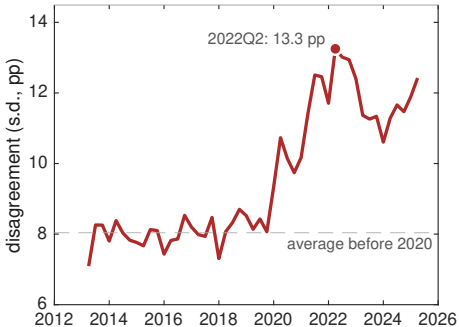
Belief Disagreement and the Housing Credit Channel

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Midwest Macroeconomic Meetings
Texas Tech University
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Who Bears the Cost of Disagreement?



8 pp before 2020 → 13 pp at the peak;
11.6 pp on average since 2024

SCE, 2013Q2–2025Q2. S.d. of 12-month house-price point forecasts across households, in percentage points.

▸ Disagreement vs. uncertainty

The question

More house-price disagreement → who loses most—and why?

- **Beliefs**: optimists or pessimists?
- **Balance sheets**: housing wealth or leverage?
- **Borrowing room**: constrained or unconstrained?

Literature: No Clear Distributional Prediction

By beliefs

- Optimists: higher bids
  $\rightarrow$ exposure to falling prices
  (Burnside, Eichenbaum & Rebelo, 2016)
- Pessimists: higher leverage
  $\rightarrow$ downside exposure
  (Bailey, Dávila, Kuchler & Stroebel, 2019)

By income and wealth

- Housing-heavy wealth
  $\rightarrow$ larger consumption losses in a belief-driven cycle
  (Kaplan, Mitman & Violante, 2020)
- Middle/high-income borrowers
  $\rightarrow$ rising share of crisis defaults
  (Adelino, Schoar & Severino, 2016)

Different mechanisms and outcomes $\rightarrow$ who loses from greater disagreement?

Unclear who loses from a dispersion shock; not contradictory findings.

This Paper: Who Loses, and Why?

Contribution

Who? Marginal borrowers—households at the collateral constraint

Why? Endogenous credit tightening in response to belief dispersion

- Individual beliefs unobservable $\rightarrow$ lender prices **pool dispersion**
- Disagreement $\uparrow \rightarrow$ common LTV limit $\downarrow \rightarrow$ unequal consumption losses
- Impact consumption: aggregate -0.16%; households at the limit -2.85%
- 3.3% of households at the limit $\rightarrow$ 28.2% of the gross impact decline

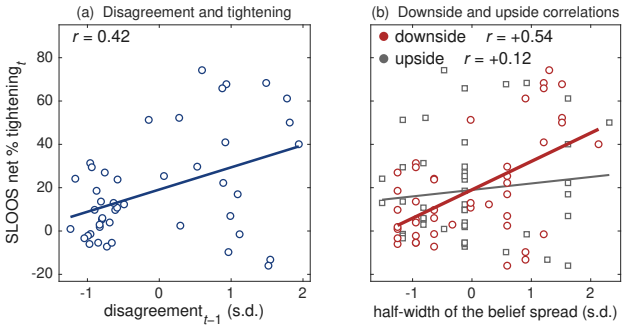
Where this paper stands

- Existing work: borrower–lender disagreement; demand-side transmission
  (Geanakoplos, 2010; Simsek, 2013; Burnside et al., 2016; Dong et al., 2025)
- Here: disagreement *within the borrower pool* $\rightarrow$ credit terms $\rightarrow$ who bears the losses

Gross decline: losses among households with falling consumption.

Motivating Facts 1–2: Disagreement and Lending Standards

Disagreement: cross-sectional s.d. of *households'* expected 12-month *home-price* growth.



Lending standards [commercial]

- Fact 1: lagged disagreement, $r = 0.42$
- Fact 2: downside $r = 0.54$; upside $r = 0.12$

Lending standards [residential]

- Similar correlation: $r = 0.43$

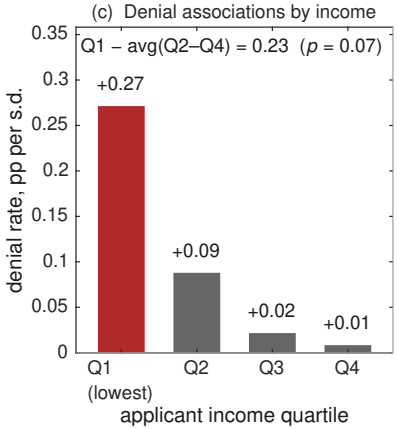
Mortgage leverage [commercial]

- -1.69 pp LTV per s.d. (s.e. 0.24)

Leverage observed where terms are negotiated: commercial mortgages

SCE/SLOOS, 2014Q1–2025Q1; both panels: commercial real-estate standards. CMBS: 10,956 loans, 2016Q4–2024Q4. Associations, not causal estimates; realized LTV ≠ credit ceiling. ▸ SLOOS ▸ CMBS ▸ Measurement

Motivating Fact 3: Denials Rise Most at Low Income



Mortgage applications [residential]

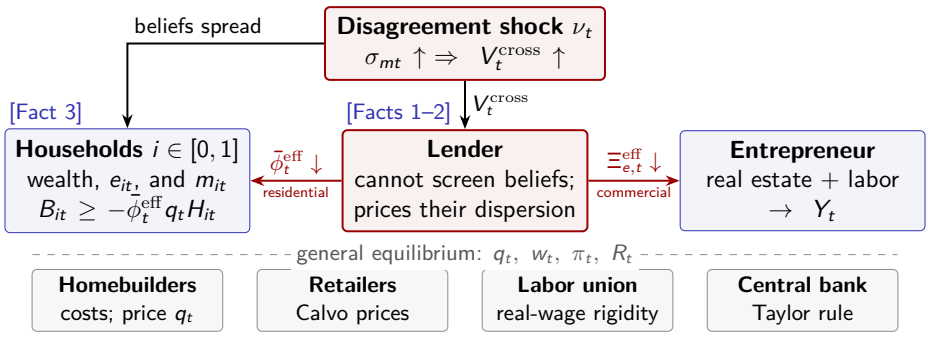
- HMDA: 30.4 million home-purchase applications
- Denial rate on state disagreement, by applicant-income quartile
- Point estimates fall with income: 0.27, 0.09, 0.02, 0.01 pp per s.d.
- Bottom quartile vs. the rest: 0.23 pp (s.e. 0.13, $p = 0.07$)

Who is denied when disagreement is higher: low-income applicants

Paper, Figure 2(c); 2018–2024. State/year FE; state-clustered s.e. Bottom vs. top: 0.26 pp, $p = 0.10$; individual coefficients insignificant at 5%. Applicant-income quartiles $\neq$ model wealth groups. ▸ HMDA details

Model: Common Tightening, Unequal Exposure

Ex-ante identical households: common preferences, β_h, income and belief processes
$\rightarrow$ borrowers, savers, and constrained households emerge endogenously



Common LTV tightening $\rightarrow$ largest losses at the constraint

One lender promise scales the household (residential) and entrepreneur (commercial real-estate) limits. Housing collateral: Iacoviello (2005). No realized default. ▸ Households ▸ Rest of economy ▸ Related work

Households: Housing Services and Collateral

- Choices: consumption C_{it}, housing H_{it}, bonds B_{it}
- Heterogeneity: wealth, productivity e_{it}, house-price belief m_{it}

Budget constraint

$$C_{it} + q_t[H_{it} - (1-\delta_h)H_{it-1}] + B_{it} = \frac{R_{t-1}B_{it-1}}{\pi_t} + w_t e_{it} L_{et} + F_t.$$

Collateral constraint

$$B_{it} \;\geq\; -\bar{\phi}_t^{\text{eff}}\, q_t H_{it}, \qquad B_{it} < 0 \text{ denotes borrowing.}$$

Same LTV limit + different states → different distances to the constraint

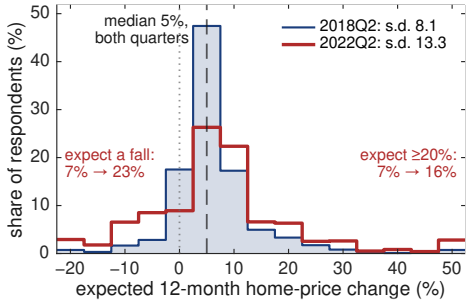
Preferences: consumption, housing services, labor disutility. F_t: rebated profits; q_t: real house price.

▸ Preferences and FOCs

Disagreement Shock: Wider Cross-Section of Beliefs

Perceived resale value: $m_{it}\mathbb{E}_t[q_{t+1}], \quad 0 < m_{it} < 1$

$$
\begin{aligned}
m_{it} &= \frac{1}{1+e^{-z_{it}}}, \\
z_{it} &= (1-\rho_m)\bar{z} + \rho_m z_{it-1} + \sigma_{mt}\varepsilon^m_{it}, \\
\log\sigma_{mt} &= (1-\rho_{sm})\log\bar{\sigma}_m + \rho_{sm}\log\sigma_{mt-1} + \sigma_\nu \nu_t, \\
V^{\text{cross}}_t &= \rho^2_m V^{\text{cross}}_{t-1} + \sigma^2_{mt}.
\end{aligned}
$$



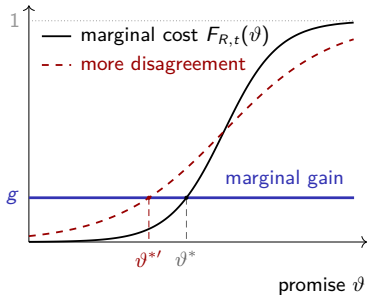
In the data: same median, wider spread

- $\nu_t > 0 \;\rightarrow\; \sigma_{mt} \uparrow \rightarrow\; V^{\text{cross}}_t \uparrow$; mean z unchanged; negligible change in mean m
- Dispersion accumulates: $\rho^2_m \approx 0.85$ of V^{cross}_{t-1} carried forward $\rightarrow$ gradual build-up, slow decay
- $m_{it} < 1 \;\rightarrow$ discount on resale *level*, including steady state

Histogram: SCE 12-month house-price point forecasts, 2018Q2 vs. 2022Q2; end bins hold winsorized mass. Cognitive discounting motivated by Gabaix (2020); time-varying dispersion process as in Fernández-Villaverde et al. (2015).

Lender: The Maximum Promise

- Individual beliefs unobservable at origination → common underwriting terms
- Promise ϑ; capped payoff $\min\{\vartheta, \mathcal{R}_{t+1}\}$ per unit of collateral; margin $g = 1 - R^f/R^L$



$$\max_{0\le\vartheta\le 1} \underbrace{g\vartheta}_{\text{margin income}} - \underbrace{\mathbb{E}_t^{\text{lend}}[(\vartheta - \mathcal{R}_{t+1})_+]}_{\text{expected credit loss}}$$

Marginal gain = marginal expected loss: $F_{R,t}(\vartheta_t^*) = g$

- Small g → low recovery quantile
- Wider downside tail → same shortfall probability at a smaller promise

Downside exposure ↑ → borrowing capacity ↓ [Fact 2]

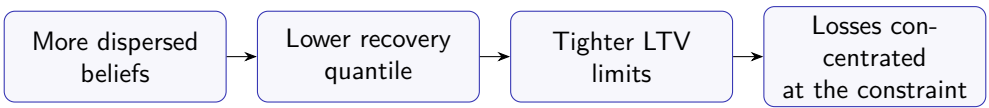
Merton (1974); Freimer & Gordon (1965); Jaffee & Modigliani (1969). Schematic curves; g enlarged for visibility (calibration: 1%). No realized equilibrium default. ▸ Derivation and loss cap

From Disagreement to Borrowing Capacity

Recovery: workout at the borrower's valuation, or forced sale to another buyer from the pool
→ recovery drawn from the distribution of beliefs

$$\operatorname{logit} \mathcal{R}_{t+1} = \bar{z} + \kappa_R \sqrt{V_t^{\text{cross}}}\, \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim N(0,1).$$

$$\operatorname{logit} \vartheta_t^* = \bar{z} - \chi_q \sqrt{V_t^{\text{cross}}}, \qquad \bar{\phi}_t^{\text{eff}} = \bar{\phi} \frac{\vartheta_t^*}{\vartheta_{ss}^*}, \qquad \chi_q = \kappa_R \Phi^{-1}(1-g) > 0.$$



- Same scaling $\vartheta_t^*/\vartheta_{ss}^*$ → entrepreneur's LTV limit
- Equilibrium: LTV rule; borrowing at R_t; no realized default

Paper, Section 3.3. Belief dispersion–recovery risk link: maintained modeling assumption. ▸ Rule in pictures

Calibration and Solution

Credit sensitivity χ_q: from lending standards

$$S_t = \alpha + \sum_{\ell=0}^{2} \beta_\ell D_{t-\ell} + \sum_{\ell=1}^{2} \gamma_\ell S_{t-\ell} + \epsilon_t$$

- S_t: net % tightening standards [commercial]
- D_t: disagreement (s.d.); $\hat{\beta}_1 = 9.54$ on D_{t-1} [Fact 1]
- 1 pp net tightening $\equiv$ 1% lower credit limit: $-\hat{\beta}_1/100 = -\chi_q(1-\vartheta^*_{ss})s_V$
- $\chi_q = 8.59$; $\hat{\beta}_1 \pm 1$ s.e.: $\chi_q = 4.15,\ 11.79$

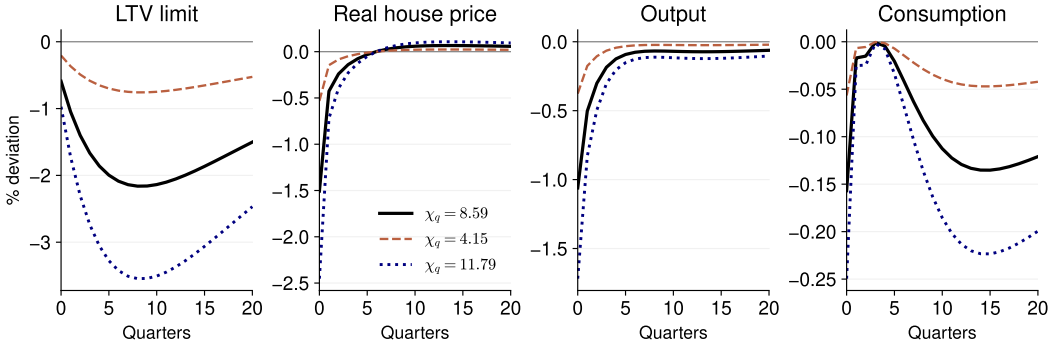
Selected parameters

Household LTV $\bar{\phi}$	0.80
Mean belief parameter $\bar{m}$	0.75
Belief persistence ρ_m	0.92
Dispersion persistence ρ_{sm}	0.942
Innovation scale σ_ν	0.066

Second-order perturbation + pruning; 72 household bins

SLOOS, 2014Q1–2025Q1. Right-hand side: $d\log\bar{\phi}_t^{\text{eff}}/d\widetilde{D}_t$; $\widetilde{D}_t$ standardizes $\sqrt{V_t^{\text{cross}}}$; s_V is its simulated time-series s.d. Mapping: calibration assumption. Locally fixed binding status. ▸ SLOOS estimates ▸ Details ▸ Numerics ▸ Model fit

Small Aggregate Response, Persistent Credit Tightening

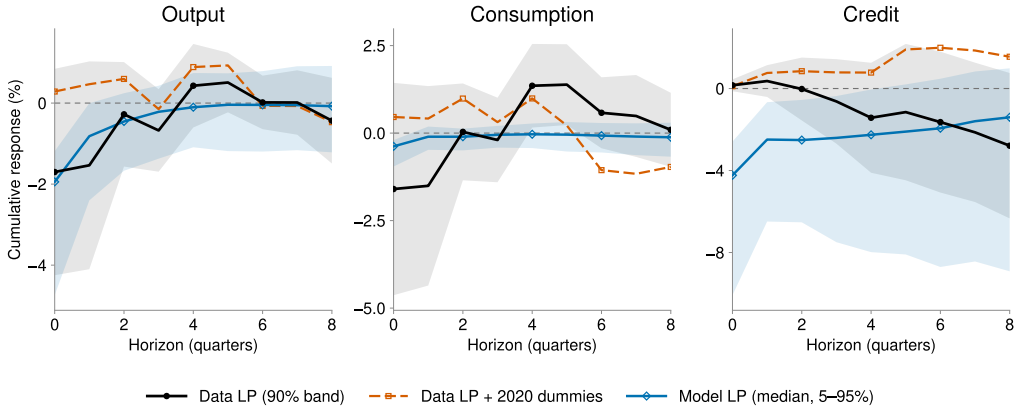


- LTV limit: -0.59% on impact $\rightarrow$ -2.16% in quarter 8; dispersion accumulates
- Lender pricing off ($\chi_q = 0$): output response $< 0.001\%$ $\rightarrow$ credit channel = transmission

Disagreement shock $\rightarrow$ $Y_0 \approx -1\%$, $C_0 \approx -0.16\%$

One-s.d. shock; percent deviations from no-shock paths. Lines: calibrated credit sensitivities, not a confidence band.
Paper, Figure 3; shutdown: Section 3.1. Measured path

Aggregate Dynamics: Model and Data

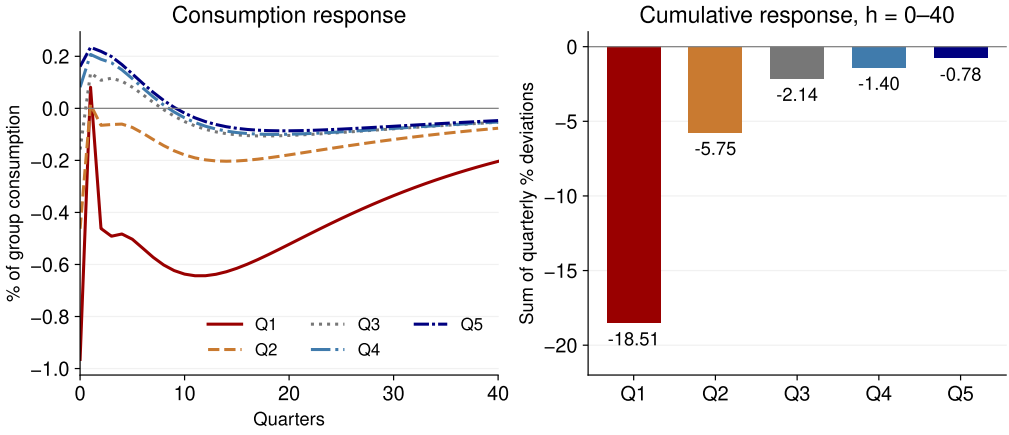


Baseline: initial contraction; sustained credit weakness

Same LP specification on observed and simulated data; disagreement levels, not identified shocks. Wide bands; sensitivity to COVID controls. Credit (data): loans secured by commercial real estate. Paper, Figure 4.

▸ Full figure and method

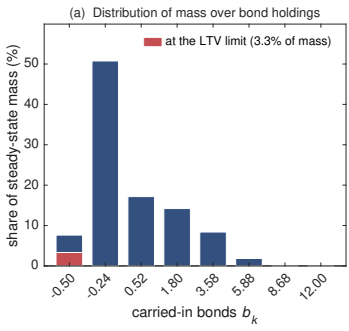
Consumption Losses by Wealth Group

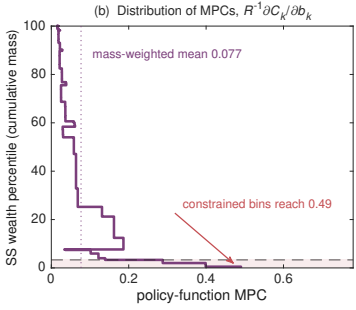


Cumulative consumption: Q1 −18.51 vs. Q5 −0.78

Units: sum of quarterly % deviations, $h = 0, \ldots, 40$; not permanent losses or welfare equivalents. Q1–Q5: paper's bond-wealth groups; unequal population shares. Mass: Q1 11.65%; Q5 21.75%. ▸ Exact quintiles ▸ Housing and net worth

Who Is at the Constraint? Mass and MPCs





- At the LTV limit: 3.3% of households
- Little borrowing room → high MPCs

High MPCs + limited borrowing room → large spending response

Paper, Figure 7(a)–(b). MPCs are finite differences on the bond grid at fixed income and belief states; estimates near the constraint require care. ▸ Full household results

Marginal Borrowers: Concentrated Losses

Concentrated exposure

3.3%

of households at the limit

⇓

28.2%

of the gross impact consumption decline

Impact consumption response by channel (%)

	Constrained borrowers	Unconstrained borrowers
Credit limit	−0.40	−0.13
Labor income	−1.28	−0.43
House price	−1.42	−0.05
Real rate	−0.18	+0.10
Transfers	+0.47	+0.21
All channels	−2.80	−0.30

Credit tightening → lower labor income and collateral values → larger losses

Paper, Table 6: local channel decomposition, percent of group consumption; rounded rows. Total simulated constrained impact is −2.85% (Table 7). Gross-decline share excludes households whose consumption rises. Lender pricing off: output response $< 0.001\%$. ▸ Mass and MPCs

Income and Beliefs: Who Is Most Exposed?

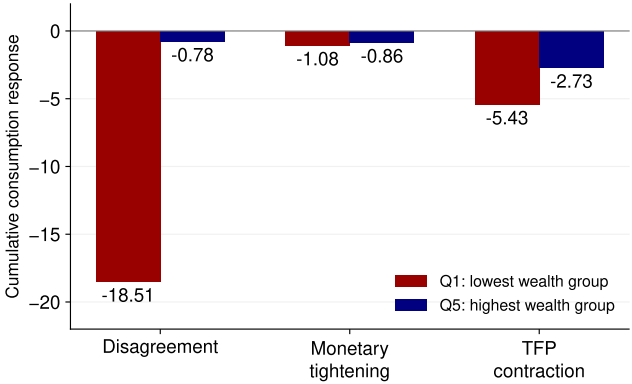
	Pessimistic	Middle belief	Optimistic
Cumulative consumption response, $h = 0, \ldots, 40$			
Low income	−6.5	−10.5	−19.3
Middle income	−1.5	−2.3	−3.8
High income	−0.6	−0.8	−1.0
Low-income share at the limit	8.0%	12.2%	20.7%

- Constrained bins: all in the low-income state [Fact 3]
- Within low income: relative optimism → higher constrained share

Downside tail → credit terms; borrowing room → who loses most

Paper, Table 9. Percent of each cell's own steady-state consumption; fixed mass weights. Belief states $m = 0.712, 0.750, 0.784$ are relative optimism within the model.

Do Other Shocks Produce Such Unequal Losses?



Normalization: same aggregate impact, $\Delta C_0 = -0.16\%$

Easing that offsets ΔC_0 (≈ 8 bp): restores $\approx 25\%$ of the aggregate cumulative loss, but only $\approx 3\%$ at the limit and $\approx 6\%$ for Q1

Ten-year variance from disagreement: 92.9% of the Q1–Q5 consumption gap; 45.9% of aggregate consumption

Disagreement → losses fall more heavily on low-wealth households

Paper, Table 7. Cumulative sums, $h = 0, \ldots, 40$, in each group's quarterly consumption units; fixed steady-state weights. Scaling, the TFP sign reversal, and the easing offset (impact only; linear superposition) are local approximations. Variance: shares among three model shocks (Table 7B). Q1–Q5: paper's bond-wealth groups; unequal population shares. ▸ Income risk ▸ Variance shares

Persistent Tightening Magnifies Marginal Borrowers' Losses

Comparison: average pessimism vs. disagreement; same initial LTV tightening (-0.59%)

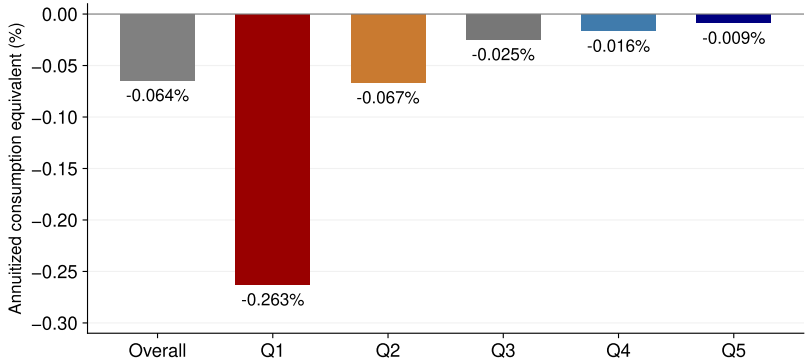
	Disagreement (benchmark)	Average pessimism
Largest LTV tightening (%; quarter)	-2.16 (8)	-0.59 (0)
Cumulative LTV response	-55.33	-9.25
Cumulative constrained consumption	-48.30	-7.89
Constrained share of cumulative gross decline (%)	27.7	28.2

Same concentration at the constraint (cumulative shares $\approx 28\%$); different depth and duration

Deeper, longer credit squeeze $\rightarrow$ about $6\times$ cumulative constrained losses

Paper, Table 8. Cumulative responses: sums of quarterly % deviations, $h = 0, \ldots, 40$; consumption relative to the group's own steady-state level. Shares refer to gross declines. Initial LTV tightening matched; aggregate consumption impacts differ. ▸ Full comparison

Welfare Losses Are Largest among Low-Wealth Households



- Utility: consumption + housing services − labor disutility
- Annuitized welfare loss: Q1 0.263% vs. Q5 0.009%; larger Q1 losses in high-rate and recession states

Paper, Table 5, Panel B: steady-state benchmark, 1,600 quarters, $\beta_h = 0.99$. Negative consumption equivalents indicate welfare losses. Q1–Q5: paper's bond-wealth groups; unequal population shares. ▸ Initial-state comparisons

Conclusion: Who Loses, and Why?

- **Beliefs?** Downside tail $\rightarrow$ credit terms;
  within low income, relative optimists lose most (-19.3 vs. -6.5)
- **Balance sheets?** Largest losses at low wealth: Q1 -18.51 vs. Q5 -0.78
- **Borrowing room?** Decisive: 3.3% at the limit $\rightarrow$ 28.2% of the gross impact decline

Why: dispersed borrower beliefs $\rightarrow$ common, persistent LTV tightening
$\rightarrow$ lower labor income and collateral prices $\rightarrow$ losses on marginal borrowers

The main message

Small aggregate consumption response; large losses at the borrowing limit

▸ Backup slides

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Backup Slides

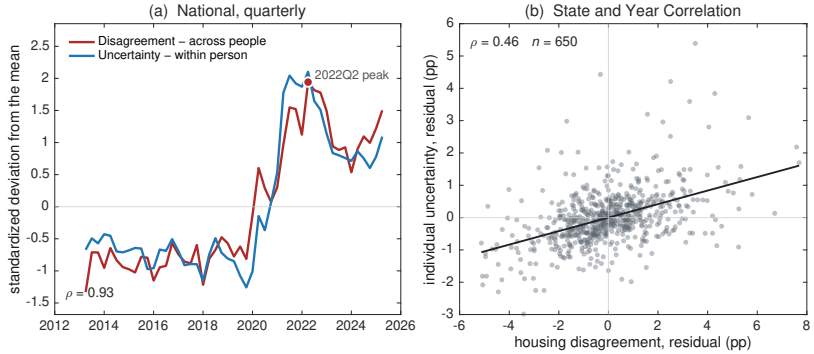
Evidence and calibration

- Disagreement vs. uncertainty
- SLOOS robustness
- Fact 1 over time
- CMBS robustness
- HMDA denials
- Michigan survey
- Calibration
- Untargeted moments
- More moments
- Local projections
- Historical path
- Back

Mechanism and household effects

- Household problem
- Rest of economy
- Lender derivation
- Recovery picture
- Mass and MPCs
- Variance shares
- Exact quintiles
- Welfare by state
- Housing and net worth
- Shock comparisons
- Income risk
- Related literature
- Numerics
- Perturbation order

Disagreement vs. Subjective Uncertainty



- **Disagreement**: s.d. of 12-month point forecasts *across respondents*
- **Uncertainty**: average IQR *within respondents' densities*
- $\rho = 0.93$ nationally; state/year FE $\rightarrow$ $\rho = 0.46$

SCE: 179,946 person-waves, 2013Q2–2025Q2; point forecasts winsorized at 1/99. Paper, Figure 1. ▸ Michigan

◂ Back ◂ Back

Lending Standards: Correlations and Conditional Evidence

	r	$t(r)$	$\hat{\beta}_1$	$t(\hat{\beta}_1)$
CRE lending standards	0.42	3.03	9.54	1.54
Residential, GSE-eligible	0.43	2.92	13.35	1.18
Downside half	0.54	4.24	1.34	0.25
Upside half	0.12	0.80	−6.38	−1.44
Subjective uncertainty	0.20	1.33	6.99	0.65
Mean forecast	−0.75	−7.44	1.11	0.25
Stock-price placebo	−0.06	−0.38	7.96	2.08

$$S_t = \alpha + \sum_{\ell=0}^{2} \beta_\ell D_{t-\ell} + \sum_{\ell=1}^{2} \gamma_\ell S_{t-\ell} + \epsilon_t.$$

- CRE coefficient excluding 2020: 7.49 ($t = 1.17$).
- Michigan: aggregate correlation 0.38; conditional lag coefficient 0.24 ($t = 0.06$).
- Michigan downside coefficient: negative; ex-2020 → significant

Paper, Tables 1 and Appendix.3. Correlations, not identified credit-supply shocks.

◂ Back ◂ Back ▸ Over time

Commercial Mortgage Leverage: Specification Sensitivity

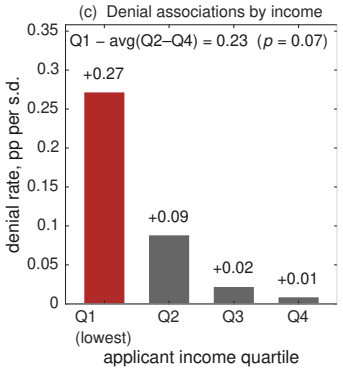
Specification	LTV change (pp)	NW(4) s.e.	Loans
Disagreement alone	−1.686	0.237	10,956
Add subjective uncertainty	−2.029	0.659	10,956
Add mean expectation	−1.672	0.213	10,956
Add both	−0.838	2.179	10,956
Add lagged LTV	−1.205	0.400	10,514
Quarterly first differences	−0.009	0.650	10,514
Four-quarter differences	−1.772	0.339	9,110
Michigan disagreement	−1.486	0.259	10,956

- Securitization LTV; property/state/appraisal adjustment; origination-quarter groups
- Joint belief controls → insignificant; quarterly differences → near zero
- Realized leverage: borrower choices + securitization selection

Paper, Table 2; 2016Q4–2024Q4.

Back

Residential Mortgage Denials: Sensitivity by Applicant Income



HMDA: 30.4 million applications, 2018–2024

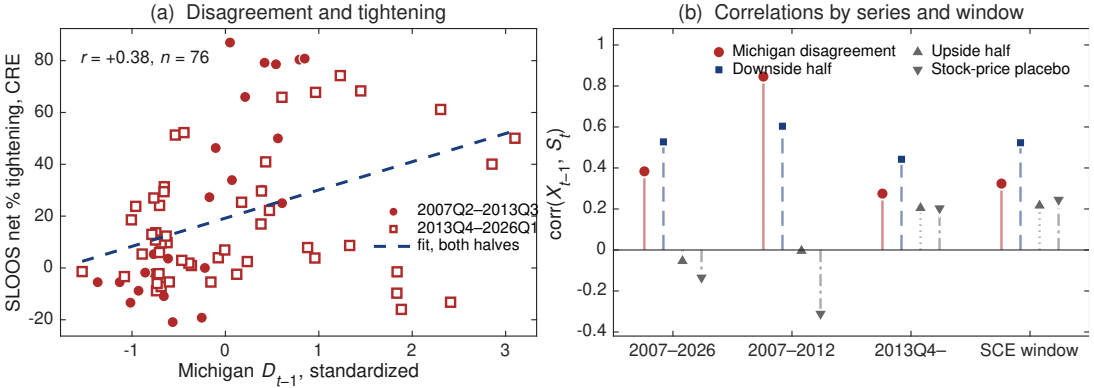
$$\text{Denial}^{q}_{s,y} = \alpha^{q}_{s} + \delta^{q}_{y} + \beta^{q} D_{s,y} + u^{q}_{s,y}$$

- Quartiles within state-year; state/year FE; clustered s.e.
- Point estimates: 0.27, 0.09, 0.02, 0.01 pp
- Bottom vs. the rest: 0.23 pp (s.e. 0.13, $p = 0.07$)
- Bottom vs. top: 0.26 pp, $p = 0.10$
- Respondent-weighted: 0.35 pp, $p = 0.055$

Paper, Figure 2(c), with the bottom-vs.-rest contrast as annotation. First-lien, owner-occupied home purchases; associations per s.d. of state disagreement. Individual coefficients insignificant at 5%. Bottom vs. the rest: stacked regression, bottom quartile minus the average of the other three. Weighted: state-year cells weighted by SCE respondents (replication output, `hmda_quartile_tests.csv`). Applicant-income quartiles $\neq$ model wealth groups.

◂ Back

Alternative Survey: Similar Correlation Patterns



- Michigan: local prices; SCE: nationwide prices
- Disagreement/standards correlation: 0.38; downside 0.53, upside −0.05.

Paper, Appendix Figure 2 and Table Appendix.3. Michigan correlation sample: 2007Q2–2026Q1.

Back

Household Problem and the Housing Euler Equation

$$\max \ \mathbb{E}_0 \sum_{t \geq 0} \beta_h^t \left[\frac{C_{it}^{1-\sigma_c}}{1-\sigma_c} + j \frac{H_{it}^{1-\sigma_h}}{1-\sigma_h} - \psi \frac{l_{it}^{1+\xi}}{1+\xi} \right]$$

$$C_{it} + q_t[H_{it} - (1-\delta_h)H_{it-1}] + B_{it} = \frac{R_{t-1} B_{it-1}}{\pi_t} + w_t e_{it} L_{et} + F_t, \quad B_{it} \geq -\bar{\phi}_t^{\text{eff}} q_t H_{it}.$$

$$C_{it}^{-\sigma_c} = \lambda_{it},$$

$$\lambda_{it} = \beta_h \mathbb{E}_t[(R_t/\pi_{t+1})\lambda_{it+1}] + \varphi_{it},$$

$$jH_{it}^{-\sigma_h} + \beta_h m_{it} \mathbb{E}_t[\lambda_{it+1}(1-\delta_h)q_{t+1}] + \varphi_{it}\bar{\phi}_t^{\text{eff}} q_t = \lambda_{it} q_t.$$

$$\varphi_{it} \geq 0, \qquad \varphi_{it}(B_{it} + \bar{\phi}_t^{\text{eff}} q_t H_{it}) = 0.$$

Paper, Section 3.1. Productivity follows a log AR(1), discretized by Rouwenhorst; the union sets a common workweek.

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Entrepreneur, Housing Supply, and Monetary Policy

$$
\begin{aligned}
Y_t &= A_t H_{et-1}^{\alpha} L_{et}^{1-\alpha}, \\
\log A_t &= \rho_A \log A_{t-1} + \zeta_A(\log Y_{t-1} - \log \bar{Y}) + \sigma_A \varepsilon_t^A, \\
B_{et} &\geq -\Xi_e \frac{\vartheta_t^*}{\vartheta_{ss}^*}(1-\delta_h) q_t H_{et} \frac{\mathbb{E}_t \pi_{t+1}}{R_t}, \\
q_t &= 1 + \chi_h \frac{I_t}{H_t} + \chi_{\Delta I} \frac{I_t - I_{t-1}}{H_t} - \beta_h \chi_{\Delta I} \mathbb{E}_t \left[\frac{I_{t+1} - I_t}{H_{t+1}} \right], \\
\log w_t &= \gamma_w \log w_{t-1} + (1-\gamma_w) \log w_t^{\text{flex}}, \\
R_t &= R_{t-1}^{\rho_R} \left[R(Y_t/Y)^{\phi_Y} (\pi_t/\pi)^{\phi_\pi} \right]^{1-\rho_R} \exp(\varepsilon_t^R).
\end{aligned}
$$

- Less patient entrepreneur, $\beta_e < \beta_h \ \rightarrow$ borrowing at the limit
- Calvo retailers; rebated profits; goods/housing/labor/bond clearing

Paper, Sections 3.4–3.5.

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The Optimal Promise Is a Recovery Quantile

$$L_t(\vartheta) = \mathbb{E}_t[(\vartheta - \mathcal{R})_+] = \int_0^{\vartheta} (\vartheta - r) f_{R,t}(r)\, dr, \qquad L_t'(\vartheta) = F_{R,t}(\vartheta).$$

$$\frac{d}{d\vartheta}[g\vartheta - L_t(\vartheta)] = g - F_{R,t}(\vartheta), \qquad \frac{d^2}{d\vartheta^2}[g\vartheta - L_t(\vartheta)] = -f_{R,t}(\vartheta) < 0.$$

- Expected-loss cap $L_t(\vartheta) \le \bar{\ell}$; multiplier η_t:

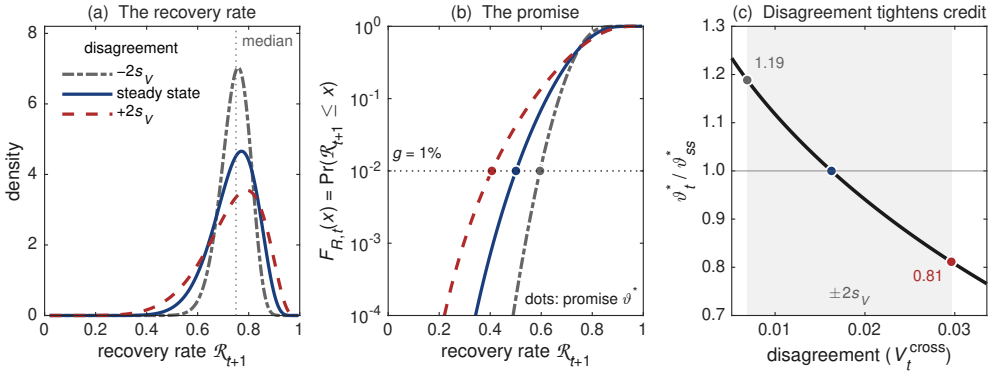
$$F_{R,t}(\vartheta_t^*) = \frac{g}{1+\eta_t}, \qquad \text{logit}\, \vartheta_t^* = \bar{z} - \kappa_R \Phi^{-1}\left(1 - \frac{g}{1+\eta_t}\right) \sqrt{V_t^{\text{cross}}}.$$

- Fixed $g < 1/2$, η_t: logit dispersion $\uparrow \rightarrow$ quantile $\downarrow$
- Fixed logit center/median $\not\Rightarrow$ fixed mean recovery in levels

Paper, lender appendix. No endogenous default or loan spread in the macro equilibrium.

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The Lender's Rule in Pictures



- Dispersion ↑ at fixed median → lower recovery quantile
- $\vartheta_t^*/\vartheta_{ss}^*$ ↓ → both household and entrepreneur limits ↓

Paper, Appendix Figure 1. At $g = 1\%$, $\kappa_R = 3.69$ and $\chi_q = 8.59$.

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Selected Calibration and the SLOOS Mapping

Block	Parameters	Values
Households	$\beta_h, j, \sigma_c, \sigma_h, \xi$	$0.99, 0.10, 2, 3, 2$
Income	ρ_e, σ_e	$0.95, 0.08$
Entrepreneur	$\beta_e, \sigma_{ce}, \alpha, \Xi_e$	$0.98, 2, 0.33, 0.10$
Prices/wages	γ, θ, γ_w	$0.75, 5, 0.90$
Housing	$\chi_h, \chi_{\Delta I}, \delta_h$	$10, 3.5, 0.025$
Beliefs	$\bar{m}, \rho_m, \bar{\sigma}_m$	$0.75, 0.92, 0.05$
Disagreement	$\rho_{sm}, \sigma_\nu, \chi_q$	$0.942, 0.066, 8.59$
TFP	$\rho_A, \sigma_A, \zeta_A$	$0.90, 0.003, 0.05$
Taylor rule	$\rho_R, \phi_Y, \phi_\pi, \sigma_R$	$0.84, 0.12, 1.50, 0.00125$

$$\widetilde{D}_t = \frac{\sqrt{V_t^{\text{cross}}} - \sqrt{V_{ss}^{\text{cross}}}}{s_V}, \qquad \frac{9.54}{100} = \chi_q(1 - \vartheta_{ss}^*)s_V.$$

Calibration: SLOOS net tightening $\rightarrow$ log credit-limit change. CMBS: supporting evidence, not target.

Paper, Table 3 and Section 4.2.

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Untargeted Moments: Model vs. Data

Selected moment	Data	Model	[5–95%]
Housing–credit–consumption comovement			
House price with consumption	0.55	0.49	[0.29, 0.64]
House price with real-estate credit	0.25	0.30	[0.12, 0.50]
Consumption with real-estate credit	0.30	0.31	[0.03, 0.56]
Relative volatility			
Investment volatility / output volatility	4.75	4.70	[4.47, 4.98]
Distributional comovement: bottom-50% net-worth share			
Consumption with the wealth share	0.33	0.25	[−0.02, 0.48]
Output leads the wealth share, one quarter	0.39	0.39	[0.20, 0.55]

Untargeted fit → housing–credit comovement and wealth-share dynamics

Paper, Table 4. Data: 1987Q1–2019Q4; wealth shares from 1989Q3. HP filter, $\lambda = 1600$. Model: median and 5–95% range across 1,000 simulated 132-quarter samples. ▸ Additional moments ◂ Back

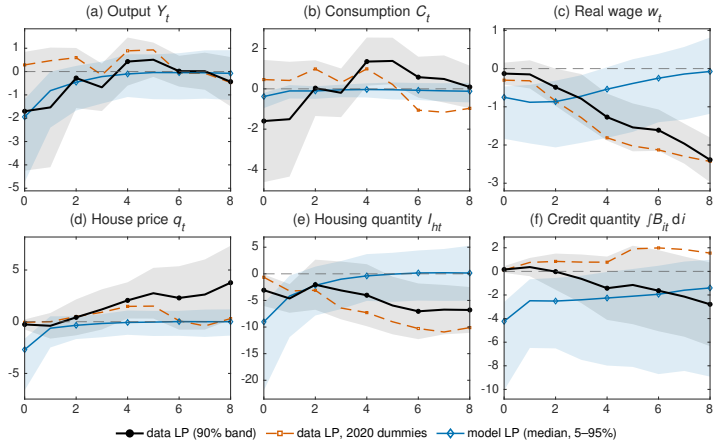
Untargeted Moments: Additional Statistics

Moment	Data	Model median
Output volatility (%)	1.04	1.30
Consumption volatility (%)	0.65	0.45
House-price volatility (%)	3.02	1.72
Investment volatility / output volatility	4.75	4.70
Credit volatility (%)	2.41	1.71
Correlation: house prices and consumption	0.55	0.49
Correlation: consumption and credit	0.30	0.31
One-quarter credit autocorrelation	0.90	0.76
Correlation: consumption and bottom-50% net-worth share	0.33	0.25

Paper, aggregate-moments table. Data: 1987Q1–2019Q4 (wealth shares from 1989Q3); HP filter, $\lambda = 1600$. Model: median of 1,000 samples of 132 quarters.

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Local Projections: Model and Data Associations

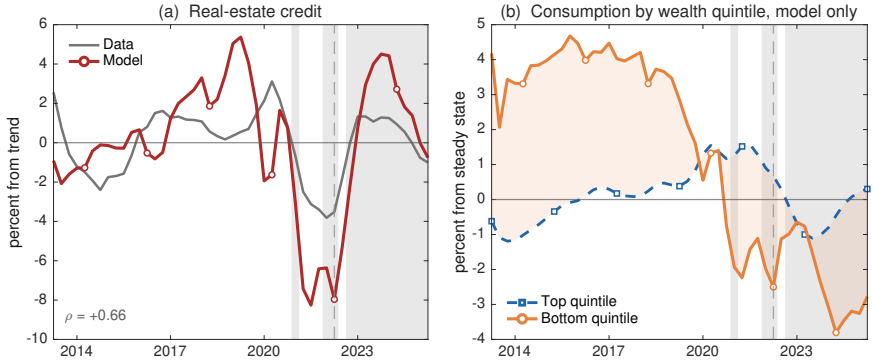


- LP: cumulative growth on standardized disagreement + two lags of disagreement and growth
- House-price response: notable miss; sensitivity to COVID controls

Paper, Figure 4. Regressor is the persistent disagreement level, not an identified structural innovation. Model: 1,000 46-quarter samples; data bands: 90% Newey–West.

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Feeding Measured Disagreement into the Model

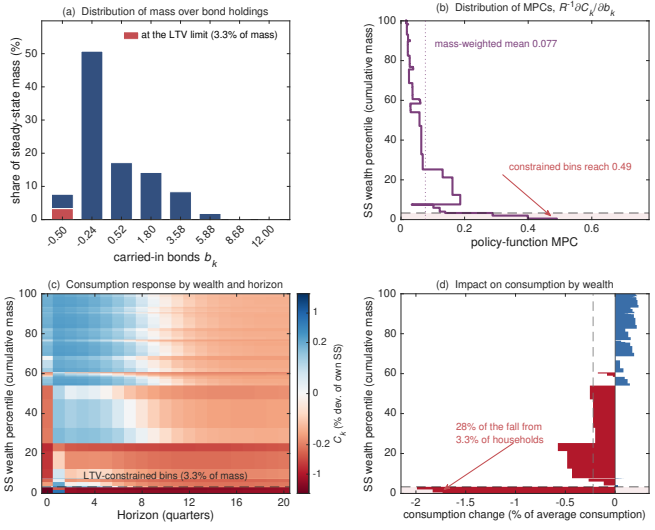


- Measured disagreement → extracted shocks; TFP/policy shocks off
- Credit comovement; COVID overshooting
- Shaded quarters → local borrowing-constraint violations

Paper, Figure 5. HP-filtered credit ($\lambda = 1600$); fixed wealth-group consumption relative to steady state.

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Mass, MPCs, and Household Consumption Losses



Paper, Figure 7. MPCs are finite differences along the bond grid at fixed income/belief states; grid estimates near a kink require care.

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The Wealth Groups Have Unequal Masses

	Q1	Q2	Q3	Q4	Q5
Paper group mass (%)	11.65	20.49	26.81	19.30	21.75
Cumulative consumption	−18.51	−5.75	−2.14	−1.40	−0.78

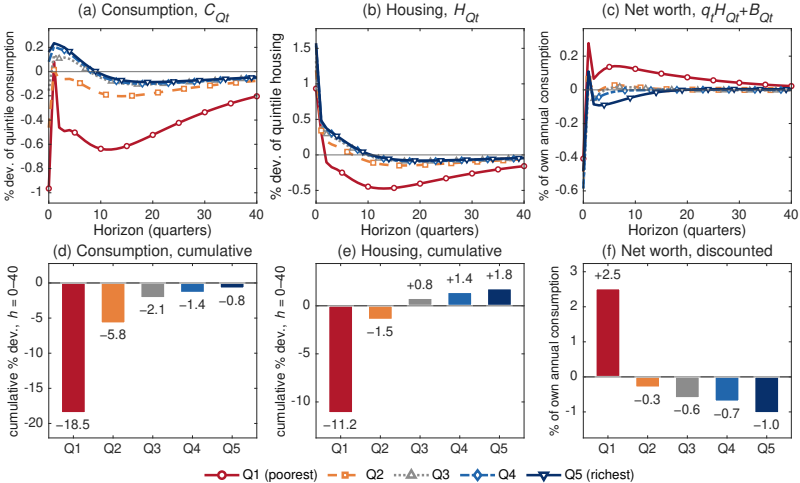
	Paper grouping	Exact 20% quintiles
Q1 cumulative consumption	−18.51	−14.08
Q5 cumulative consumption	−0.78	−0.76
Ratio of loss magnitudes	23.7	18.6

- Whole-bin assignment by cumulative bond-wealth rank → unequal group masses
- Proportional boundary-bin splits → exact quintiles; same loss ranking
- Unchanged: 3.3% constrained mass; 28.2% gross-decline share

Model_v26/Temp/2026-11-08_quintile_mass_check, independently reproducing the paper outputs. All cumulative values use $h = 0, \ldots, 40$.

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Housing and Net Worth Across Wealth Groups



Paper, Figure 6; original labels retained in this reproduction. Consumption/housing: own group levels. Net worth: percent of own annualized consumption; bottom-right panel discounted over the full path. Groups have unequal mass.

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Disagreement Explains More of Marginal Borrowers' Variation

Ten-year consumption variance share (%)	Disagreement	Monetary	TFP
All households	45.9	8.8	45.4
Constrained households	89.6	4.6	5.8
Lowest wealth group, Q1	83.6	3.8	12.6
Highest wealth group, Q5	50.3	8.4	41.3
Gap: Q1 minus Q5	92.9	2.6	4.6

$$\text{Share}_{s,H} = \frac{\sum_{h=0}^{H-1} r_{s,h}^2}{\sum_{s'} \sum_{h=0}^{H-1} r_{s',h}^2}, \qquad H = 41.$$

- IRF-based shares; three calibrated model shocks
- Model decomposition, not empirical causal attribution of inequality
- Aggregate disagreement share: 15.6% at one year; 19% in HP-filtered simulations

Paper, Table 7B and full appendix variance table; second-order solution.

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Income Risk: Who Bears the Losses?

Question: Greater income risk → the same burden at the constraint?

Impact responses	Disagreement (benchmark)	Income risk
LTV limit (%)	−0.59	0.00
Aggregate consumption (%)	−0.16	−0.28
Constrained consumption (%)	−2.85	−1.27
Constrained share of gross decline (%)	28.2	8.9

- Income risk → income transitions + precautionary saving
- Disagreement → lender-set LTV tightening → concentrated losses

Income risk: larger aggregate impact; smaller losses at the borrowing limit

Paper, Table 8. Same log-volatility innovation size/persistence; normalized aggregate labor income. Aggregate impacts differ. Three income states. Gross decline: losses among households with falling consumption.

▸ Full comparison

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Income Risk and Pessimism: Full Comparison

	Disagreement	Income risk	Average pessimism
LTV limit, impact	−0.59	0.00	−0.59
LTV limit, largest tightening (quarter)	−2.16 (8)	—	−0.59 (0)
LTV limit, cumulative	−55.33	0.00	−9.25
Aggregate consumption, impact	−0.16	−0.28	−0.06
Aggregate consumption, cumulative	−3.54	−1.89	−0.66
Constrained consumption, impact	−2.85	−1.27	−1.61
Constrained consumption, cumulative	−48.30	+2.87	−7.89
Constrained share of impact gross decline (%)	28.2	8.9	44.3
Constrained share of cumulative gross decline (%)	27.7	—	28.2
Lowest wealth group, Q1, cumulative	−18.51	−1.24	−2.85
Highest wealth group, Q5, cumulative	−0.78	−2.64	−0.23

Paper, Table 8. Responses in %; cumulative sums over $h = 0, \ldots, 40$. Groups: fixed weights, unequal sizes; aggregates include distributional movement. Income risk: matched volatility innovation. Pessimism: matched initial LTV tightening. Three-state income grid: more risk can raise expected income at the lowest state.

▸ Income risk ▸ Average pessimism

Initial States and Long-Horizon Welfare

Initial state	Overall	Q1	Q5	Q1 minus Q5
Steady-state benchmark	−0.0644	−0.2629	−0.0087	−0.2542
Pooled accepted states	−0.0580	−0.2241	−0.0108	−0.2133
Low interest rate	−0.0578	−0.2211	−0.0115	−0.2096
High interest rate	−0.0579	−0.2282	−0.0091	−0.2191
Recession	−0.0598	−0.2351	−0.0102	−0.2249
Expansion	−0.0578	−0.2221	−0.0110	−0.2111

- Annuitized CE (%): consumption, housing, and work
- High rates / recessions $\rightarrow$ larger Q1 losses and Q1–Q5 gap
- Labor $\downarrow \rightarrow$ possible impact utility gain; not long-horizon welfare

Paper, Table 5B. 1,600 quarters, $\beta_h = 0.99$; 5,000 draws per state.

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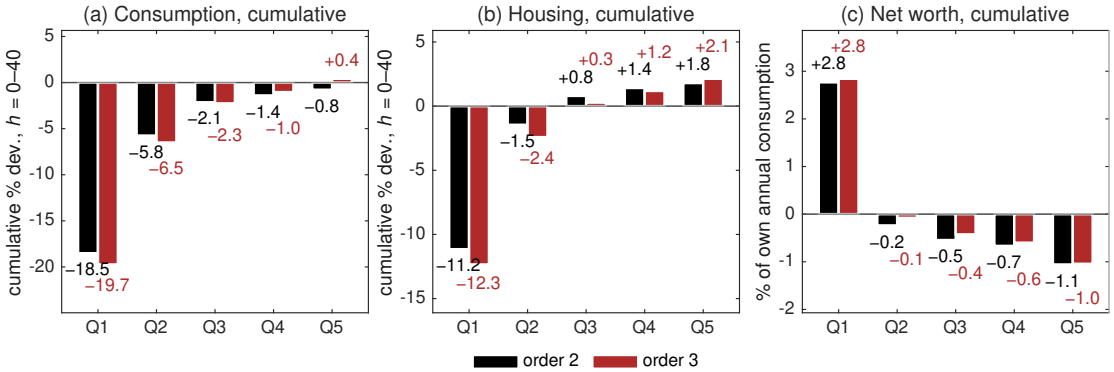
Numerical Scope of the Results

- Reiter projection/perturbation; second order with pruning.
- Grid: 8 wealth $\times$ 3 belief $\times$ 3 income states $=$ 72 household bins.
- Stationary collateral inequality; Coleman iteration, tolerance 10^{-9}
- Local perturbation: fixed binding/unconstrained split
  - Reported one-s.d. IRFs: no constraint crossings
  - Historical paths: some local violations $\rightarrow$ shaded quarters
- Second/third-order agreement on the same grid $\not\Rightarrow$ grid convergence

Paper, Section 4.1 and numerical appendix.

◂ Back ▸ Second versus third order

Distributional Results at Second and Third Order



Paper, final appendix figure. Same grid; consumption and housing cumulated over $h = 0, \ldots, 40$. Net worth here is cumulative, unlike the discounted measure in the main distribution figure.

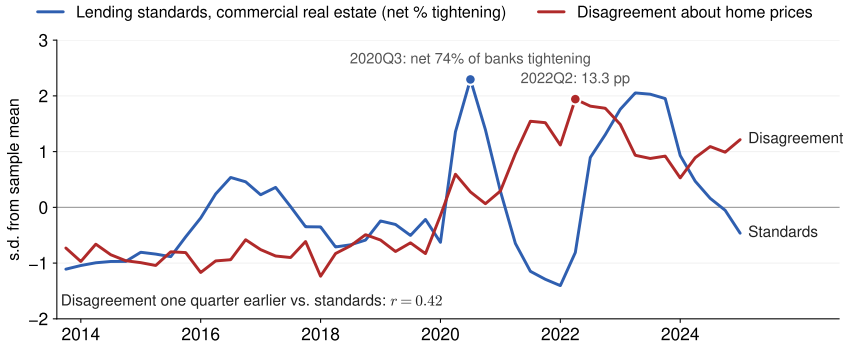
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Positioning: Who Bears the Cost of Disagreement?

Related mechanism	This paper's distinction
Borrower–lender disagreement (Geanakoplos, 2010; Simsek, 2013)	Within-pool dispersion → lender pricing
Beliefs, demand, and housing (Burnside et al., 2016; Ludwig et al., 2024)	Dispersion → tighter credit limits → constrained households
Beliefs and housing cycles (Kaplan, Mitman & Violante, 2020)	Dispersion shock; losses by wealth, income, constraint status
Heterogeneous transmission (Guerrieri & Lorenzoni, 2017; Kaplan et al., 2018)	Household effects of disagreement vs. matched monetary/TFP contractions

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Fact 1 over Time: Disagreement and Lending Standards

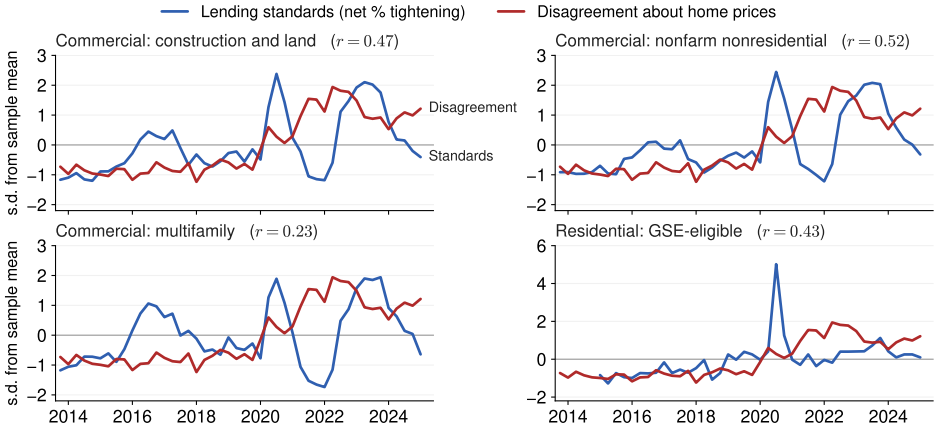


- Averages, before vs. from 2020: disagreement 8.1 → 11.5 pp; net tightening 8% → 30%
- Timing differs: 2016–17 tightening, flat disagreement; net easing into the 2022Q2 peak

Same data, sample, and lag as Fact 1: SCE/SLOOS, 2013Q4–2025Q1; average of the three commercial real-estate breakouts. Each series in s.d. from its sample mean. Associations, not causal estimates. Individual series

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Fact 1 over Time: Individual Lending-Standards Series



SLOOS net % of banks tightening, by loan category; SCE disagreement. r: disagreement one quarter earlier vs. standards, 2014Q1–2025Q1 (residential from 2015Q2). Each series in s.d. from its sample mean; the residential panel has a wider scale (2020Q3: net 55%).

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